Steane decoding and trainable quantum networks

FULL REFERENCE BUDGET | Exact CPU simulation | 3 training seeds

Problem

Can a neural network learn a Steane syndrome decoder, and can a less random quantum initialization improve its training? We compare a minimum-weight lookup, an MLP, a temporal GRU, and matched hybrid QNNs. We separately measure gradient variance versus width, depth, initialization and observable locality.

Method: four modules

1. Encode and measure. The [[7,1,3]] CSS code uses identical X/Z Hamming check matrices. Logical codewords are constructed explicitly, Pauli errors are applied, and stabilizers are projectively measured. Statevector tests verify complex logical-state recovery for every weight-one Pauli. Larger datasets use the exactly equivalent binary syndrome calculation.

2. Build trajectories. Three rounds accumulate independent depolarizing data errors with probability p per qubit per round. Readout flips each check bit with probability q. No intermediate recovery is applied. Training mixes p in {0.02, 0.06, 0.12} and q in {0, 0.03, 0.08}. Labels are minimum-weight corrections for the final true syndrome.

3. Predict and verify. Two eight-class heads predict X/Z recovery locations, including no error. Raw success requires the residual to be a stabilizer. Logical error is separately scored after a disclosed perfect final syndrome-and-lookup cleanup; zero syndrome alone is never treated as success.

4. Compare scientifically. Independent test trajectories sweep p = 0.01/0.05/0.10 and q = 0/0.03/0.08. We retain all seeds, raw gradient samples, latency, parameter counts, checkpoints, and correction statistics. A separate 64-syndrome table-fit task isolates basic supervised convergence.

Quantum architecture and controlled comparison

Six syndrome-encoded qubits pass through fixed offset rotations, three layers of RY/RZ and trainable nearest-neighbor RZZ gates, and a fixed readout basis rotation. Six Z and five ZZ expectations feed a linear head. Random angles are uniform in [-pi, pi]; mitigated angles are normal with standard deviation 0.05. A frozen small-angle circuit controls for readout-only learning. Entanglement and gradient correctness are tested.

Adam: learning rate 0.03; 120 full-batch updates; 4096 training and 1,024 validation trajectories; seeds [11, 22, 33]. Both trainable QNNs share all settings except quantum initialization. The GRU sees history; MLP/QNN see only the final observed round.

Sources: Steane (1996), Multiple-particle interference and quantum error correction; McClean et al., arXiv:1803.11173; Cerezo et al., arXiv:2001.00550. Full source links are provided in README.md.

Results: decoder training

On the complete table, mean final CE is 0.01336 for random initialization and 0.01365 for small-angle initialization. Small-angle initialization does not have lower final loss. It reaches CE <= 0.8 in 13.33 versus 19.33 mean updates (31.0% fewer). Mean table accuracies are 100.0% (small) and 100.0% (random).

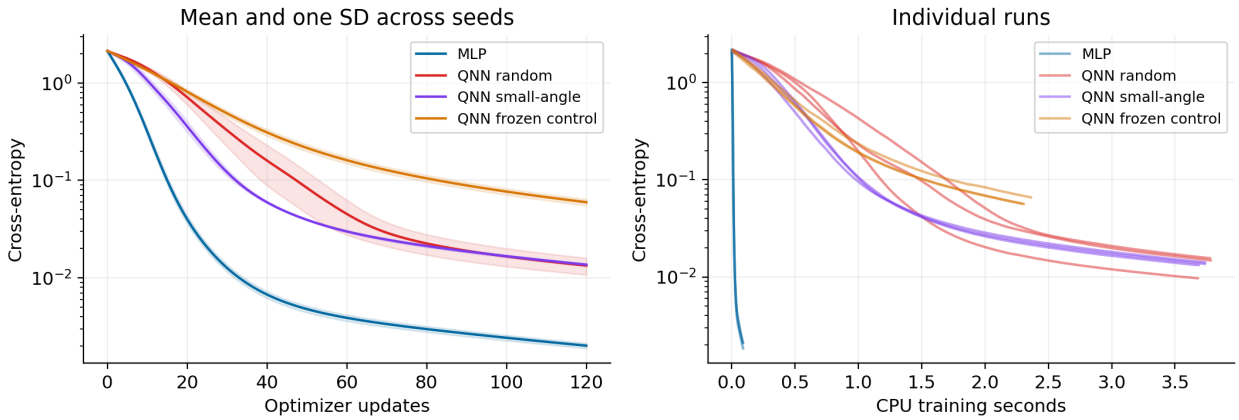


Figure 1. Complete-table learning, mean +/- one seed SD (left) and individual elapsed times (right). All 64 inputs are used for fitting; this is not a held-out generalization test.

Model	Table CE	Joint accuracy	Updates to CE <= 0.8
MLP	0.00201	100.0%	6, 7, 6
QNN random	0.01336	100.0%	22, 17, 19
QNN small	0.01365	100.0%	12, 14, 14
QNN frozen	0.05974	100.0%	20, 21, 22

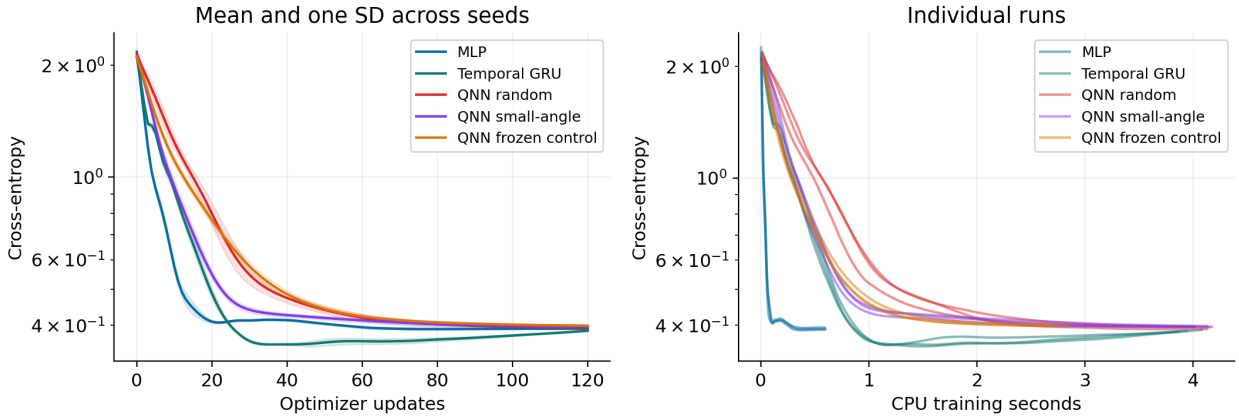


Figure 2. Independent noisy-trajectory validation. Bands show one between-seed SD, not confidence intervals. CPU elapsed time includes epoch evaluations. NR denotes threshold not reached; unsuccessful runs are retained.

The frozen control is essential: near-identity features may already preserve enough information for a linear head. Differences between trainable and frozen variants constrain how much improvement can be attributed to quantum parameter updates, rather than readout training.

Results: recovery and trainability

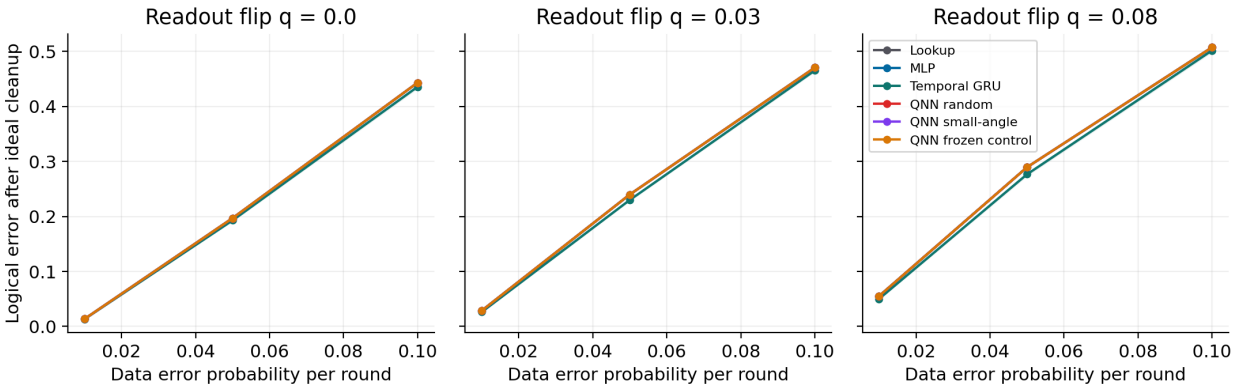


Figure 3. Logical error after ideal final cleanup; identical test samples for all models. The single-round learned curves coincide with lookup in this run. Shading is seed SD. Per-run Wilson intervals and raw failures are retained in evaluation.json. Rates include three rounds of data noise.

Model	Validation CE	Training s	Latency ms	Trainable
Lookup	--	0.0	0.005	0
MLP	0.3914	0.6	0.042	752
Temporal GRU	0.3863	4.0	0.140	2704
QNN random	0.3940	4.1	3.837	243
QNN small	0.3960	4.1	3.896	243
QNN frozen	0.3985	2.7	4.157	192

Table 1. Means over seeds; latency is the mean of single-example CPU medians. Training includes per-epoch evaluation; latency includes Python overhead. These are simulator measurements, not hardware query costs.

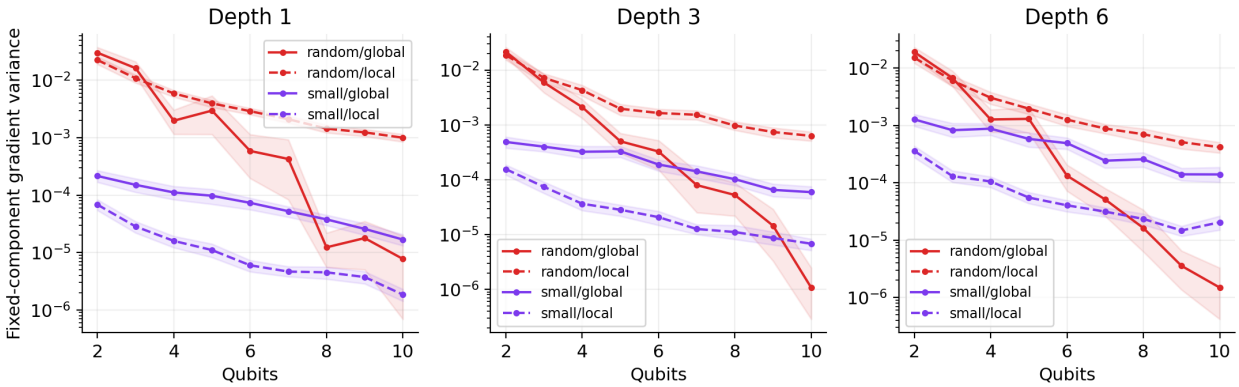


Figure 4. Fixed first-layer RY gradient component; 128 initializations per condition, depths [1, 3, 6], and bootstrap 95% intervals. Global cost is 1-P(0...0); local cost averages (1-Zi)/2. Costs here differ from decoder cross-entropy.

Mechanism and honest limits

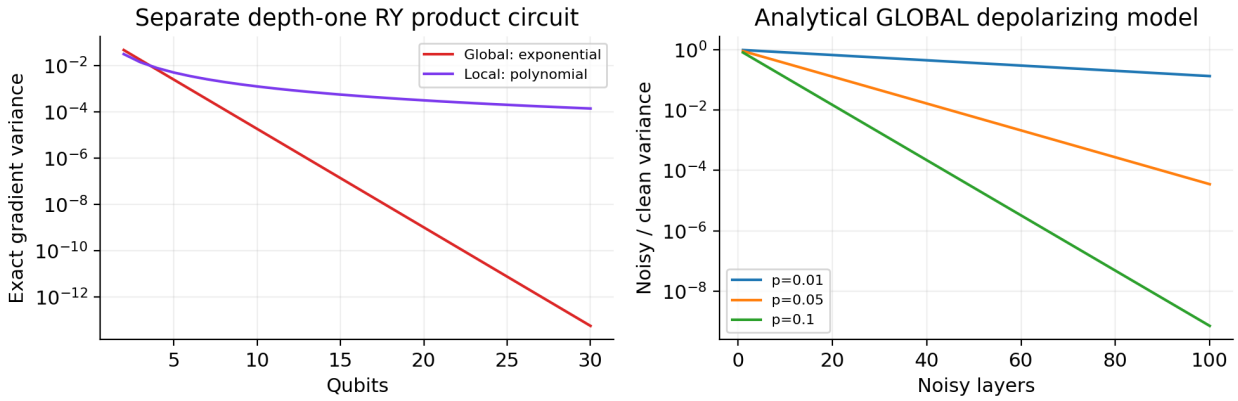


Figure 5. Separate analytical examples, not substituted measurements: shallow global-cost suppression (left) and global-depolarization contraction (right).

Why the strategies can help

Small initial angles keep the trainable circuit near identity and limit early scrambling. Local costs restrict joint dependence on all qubits. In a separate independent RY product circuit, exact gradient variances are (1/8)(3/8)^(n-1) for a global projector and 1/(8 n^2) for the mean local cost. Thus a global objective can be flat even at depth one. Local normalization itself produces polynomial decay.

For global depolarization after L layers, each gradient contracts by (1-p)^L, and its variance by (1-p)^(2L). This illustrates a noise-induced depth penalty under that restricted channel; it is not a local gate-noise simulation or a hardware error-mitigation result.

Measured finite-range slopes

Depth	Initialization	Cost	log10 variance slope / qubit
6	random	global	-0.5243
6	random	local	-0.1848
6	small	global	-0.1266
6	small	local	-0.1563

At depth 6, global-cost slopes are -0.524 for random initialization and -0.127 for small-angle initialization. Random/local gives -0.185. Both strategies slow the measured finite-range decay. These finite-range fits do not establish an asymptotic law. Polynomial decay can resemble exponential decay over a short interval; improvements need not hold at every seed or noise setting.

Limits and reproducibility

Teacher labels are minimum-weight, not noise-optimal logical cosets. The temporal model receives additional information. Independent trajectories can share syndromes; the table task is deliberate memorization. Ideal final cleanup is an extra assumption. Only three training seeds are used. No faulty ancilla extraction circuit, finite-shot quantum sampling, local quantum gate noise, or hardware execution is included. No quantum advantage is claimed.

Reproduce: Python 3.12.14, PyTorch 2.5.1+cpu, NumPy 2.1.3. Pinned requirements, a detailed README, Colab notebook, tests, raw samples and checkpoints accompany this report.